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Developer Build Spec — Shunya Protocol MVP

📝 Dev Notes & Suggestions — Shunya Protocol MVP

📋 **Project:** Shunya Protocol — Zero-Knowledge Identity Engine
Scope: 30-Day MVP Sprint (Testnet)
Classification: Confidential — Share only with contracted developer

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What We're Building

A **client-side Zero-Knowledge KYC SDK** that lets users prove identity claims (e.g., *"I am over 18"*) using their government-issued Offline Aadhaar XML — without any PII ever hitting a server.

- **Proof generation** happens locally on the user's device
- **Verification** is routed through `zkVerify` network at sub-cent cost
- **Attestation** is settled on `Base` L2 via **Ethereum Attestation Service (EAS)**

🛡️ **Core Principle:** Zero PII is stored or transmitted. All proving is client-side.

Architecture Overview

```
flowchart LR
    A["User Device\n(Aadhaar XML)"] -->|"ZK Proof"| B["zkVerify Network\n(Verificatio
n)"]
    B -->|"Attestation Receipt"| C["Base L2\n(EAS Smart Contract)"]
    C -->|"Status Flag"| D["B2B Client App\n(Access Granted)"]
```

Layer	Technology	Your Deliverable
Frontend (Client-Side Proving)	React Native / Next.js + Anon Aadhaar fork (Circom/Rust)	Modify ZK circuits, build demo UI
Middleware (Verification)	zkVerify Network (Testnet)	Build API bridge / routing logic
Backend (Settlement)	Base Sepolia (Testnet) + EAS	Deploy EAS Resolver smart contract

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MVP Deliverables (Strict Scope)

1. Smart Contracts

- Deploy an **EAS Resolver contract** on **Base Sepolia Testnet**
- The contract must **only accept** verification payloads originating from the zkVerify testnet
- Issue an on-chain attestation in the format: `Wallet Address 0x... isOver18: TRUE`

2. Cryptography (Circuit Modification)

- Fork the **Anon Aadhaar** repo (Ethereum PSE's Circom/Rust circuits)
- Modify circuits to assert **only two conditions:**
  1. `Age > 18`
  2. `Signature == Valid UIDAI Public Key`
- Keep circuit complexity minimal — no state-level restrictions in V1

3. Routing Logic (API Bridge)

- Build the middleware that:
  1. Takes the locally generated ZK proof from the client
  2. Sends it to **zkVerify testnet** for verification
  3. Receives the lightweight **Attestation Receipt**
  4. Pushes the receipt to the **Base smart contract** to trigger EAS attestation

4. Demo UI (1-Page Web App)

- A simple Next.js web app for **grant demonstrations** with three states:
  1. **"Upload XML"** — button to upload Offline Aadhaar XML (.zip)
  2. **"Generate Proof"** — loading state while client-side proving runs
  3. **"Verified on Base"** — success screen showing the on-chain attestation

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Out of Scope (Do Not Build)

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- Production mobile SDKs (iOS/Android)
- Face-matching or liveness detection
- Complex state-level location restrictions
- Consumer-facing wallet application

Key Technical References

- **Anon Aadhaar** (Circom circuits to fork) — github.com/privacy-scaling-explorations/anon-aadhaar
- **zkVerify** — zkverify.io
- **Base (Coinbase L2)** — docs.base.org
- **Ethereum Attestation Service (EAS)** — attest.sh

Performance Targets

Category	Metric	Target
⚡ Speed	Client-side proof generation time	**< 10 seconds** on a standard Android device
💸 Cost	Total gas cost per verification (zkVerify + Base)	**< $0.05 (₹4)**

Key Decisions from Discussion (30 Mar)

> **Project Name:** Shunya Protocol (शून्य = Zero in Sanskrit — representing Zero-Knowledge)

🎯 #1 Priority: Clean UX for Non-Web3 Users

- **This is NOT a hackathon project.** It must be a 100% consumer-friendly product.
- Target users are non-web3 people — if they get frustrated onboarding, the B2B clients (e-sports companies, dating apps) will reject the SDK.
- Zero Web3 jargon or friction should be visible to the end user.
- There is **no existing industry standard** for this UX — whatever we build becomes the benchmark.

🔑 User Onboarding Flow (Minimal Friction)

1. **Phone number login** (like Razorpay-style — simple OTP)
2. **Upload DigiLocker Aadhaar QR screenshot** (almost everyone in India has DigiLocker; Aadhaar data is already masked in it)
3. System extracts QR data, generates ZK proof client-side, issues on-chain attestation
4. **Done** — user never sees wallets, gas fees, or blockchain terminology

🏦 Wallet Abstraction (Invisible to User)

- Use **Coinbase Smart Accounts** — auto-generate a wallet per user internally, tied to their phone number
- User never creates or sees a wallet
- **Gas abstraction is mandatory** — Coinbase has agreed to sponsor gas costs
- Reference: Study **Privy** architecture — market standard for invisible wallet UX (Google sign-in → auto wallet creation, zero barriers)

🪪 Primary Key: Aadhaar UID (Not Phone Number)

- Aadhaar QR code contains a **UID** — use this as the primary identifier instead of phone number
- Phone number is only for login/OTP, not for identity mapping

- QR code data includes: **Name, Age/DOB, Gender, Address, (masked) Aadhaar number**
- Avoids the multi-phone-number problem entirely

📊 Data Points to Verify (MVP)

- **Age** (18+ assertion) — core use case
- **Name** — minimum identity
- **Gender** — critical for dating app vertical
- Do NOT try to verify state/location in V1 (keep circuit complexity low)

⚠️ DigiLocker — Do NOT Embed

- Embedding DigiLocker directly increases cost and defeats the purpose
- If users can just use DigiLocker for verification, there's no reason for our system
- Instead: user takes a screenshot from DigiLocker → uploads to our app → we process the QR

🔗 Tech Clarification: Anon Aadhaar ≠ zkVerify

- **Anon Aadhaar** = ZK circuit for Aadhaar proof generation (client-side)
- **zkVerify** = separate verification chain (middleware)
- These are independent — explore if alternative verification approaches exist beyond zkVerify
- ZK commitment storage needs to be handled invisibly (not like MetaMask where users manage keys locally)

🎮 Target Verticals (GTM)

1. **E-sports** (primary) — 18+ age verification required by new laws. Target companies: Nodwin Gaming, Letsplay Now, Root Mobile, BGMI tournament organizers
2. **Dating apps** — gender + age verification (minors getting in, catfishing). Newer dating apps actively looking for this solution
3. **Consumer apps** with strict age policies — social media age gates (Australia 16+, France, India upcoming within 1 year)

4. **Future expansion:** Reputation protocol (player skill verification for e-sports team segmentation), DeFi (long-term, needs proven traction first)

🏁 Competitive Landscape

- **Worldcoin** — going physical hardware route, won't work for this use case
- **Polygon ID** — exists but Polygon got distracted with DeFi/ZK EVM
- **Gitcoin** — not focused on Indian identity verification
- **No direct competitor** doing ZK-native, sub-cent Aadhaar verification for Indian market

⏰ Timeline

- **Hard deadline: April 10** (MVP must be ready to pitch)
- Developer to design system architecture first → share for feedback → then build
- Estimated build time: ~1 week + 2-3 days after architecture approval
- **Feedback loop required** — progressive reports, iterate on UX until it doesn't feel overwhelming

💡 Key Mindset Reminders

- Technology is not the issue — **presentation and consumer-friendliness is**
- Minimum friction, not zero friction (zero is impossible)
- Regulations and compliance can make or break the startup (reference: RMG industry collapse in India)
- Build a **global consumer product**, not just for India — markets are open everywhere
- Need a working MVP to pitch for grants — can't pitch just an idea

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📬 **Questions / Clarifications?** Reach out directly. All work is on testnet for now — move fast and document as you go.